

## 2.1 Use spreadsheet tools and techniques to create the solution

### 2.1.1 Data handling and manipulation

- ☐ Data validation
  - Lookup
  - Range check
  - Text length
  - Limited choice
    - Drop down lists
    - Radio buttons
    - Tick list
- ☐ Cell formatting
- ☐ Conditional formatting
- ☐ Sorting
- ☐ Filters
- ☐ Formulae
  - Operators
  - Parenthesis
- ☐ Relational operators
- ☐ Naming cells
- ☐ Cell references
  - Relative
  - Absolute
  - Named
  - Multi-sheet referencing
- ☐ Functions
- ☐ Pivot tables
- ☐ Importing different file types
- ☐ Entering different data types
- ☐ Data types
  - Boolean
  - Date
  - Time
  - Text
  - Numeric
    - Integer
    - Number/Real
    - Currency
    - Percentage
    - Decimal
- ☐ Security measures
- ☐ Modelling tools

To include:

- Create a spreadsheet solution that is fit for purpose
- Meaningful worksheet names
- Manipulation of data using formulas and functions
- Built in functions including SUM, MIN, MAX, AVERAGE, COUNT, IF, COUNTIF, LOOKUP, VLOOKUP, HLOOKUP, AND, OR, DATE, TODAY, SUMIF, SUBTOTAL
- Relational operators including =, <, >, <=, >=, <>
- Solving formula errors ( #DIV/0, #NAME?, #REF! etc)
- Effective validation checks within the spreadsheet solution
- Naming of cells or a group of cells
- Use of appropriate security measures such as lock cells, password protected workbook, worksheet editing
- Use of different cell formatting options such as alignment, border, font, shading, text wrap and currency matching the plan and design of the solution
- Modelling tools such as what-if and goal seek to predict different outcomes

## **Unit R060: Data manipulation using spreadsheets**